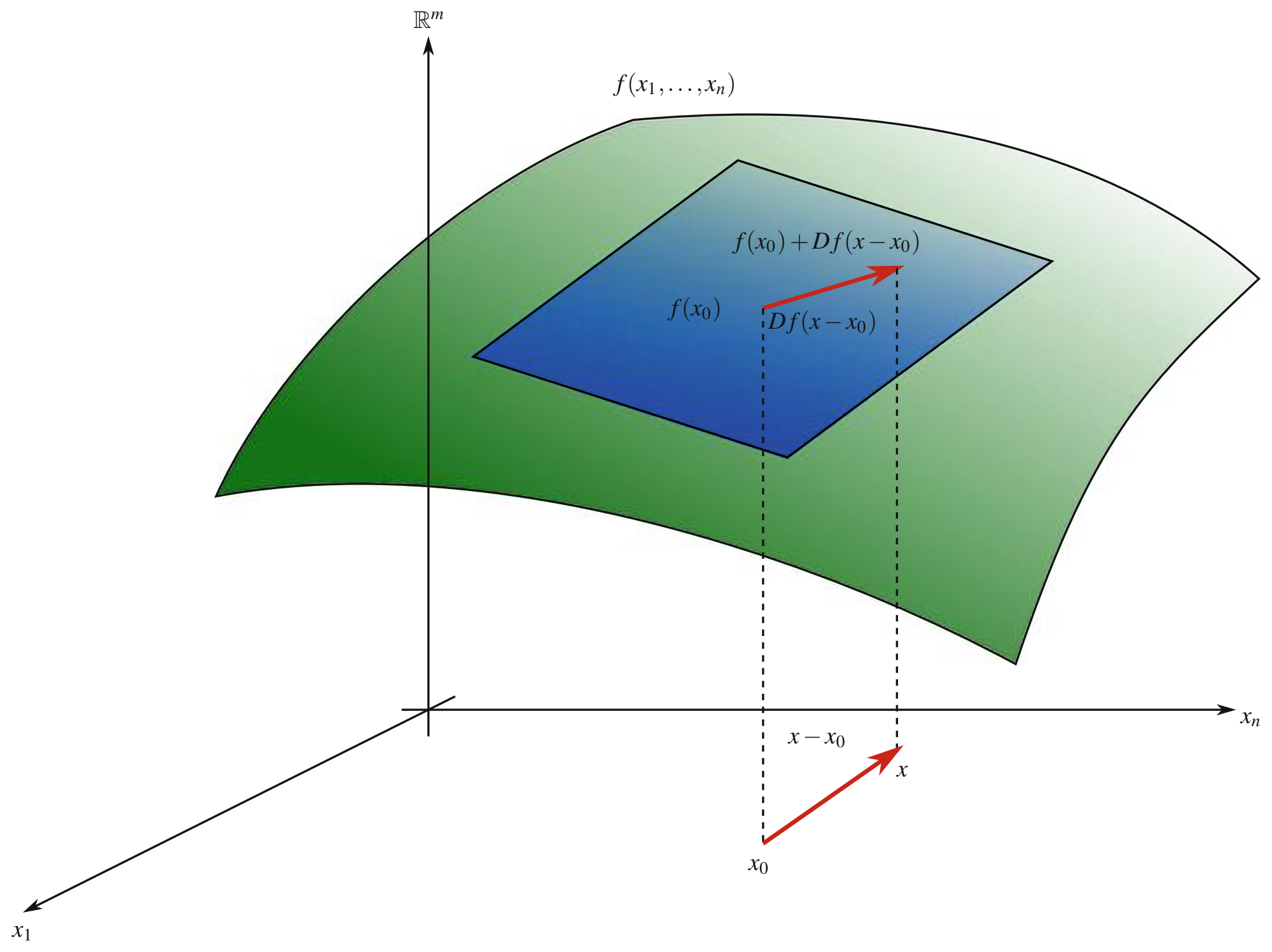

$\mathbb{R}^m$
$f(x_1,\ldots,x_n)$
$f(x_0)+Df(x-x_0)$
$f(x_0)$
$Df(x-x_0)$
x_n
$x-x_0$
x
x_0
x_1